

EAGLE

Ignacio Muñoz¹ & Jorge Dimter¹

1. Institute of Astronomy, Pontificia Universidad Católica de Chile, Santiago, Chile

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ABSTRACT

Hydrodynamical simulations have shown to be important tools in astrophysics for understanding the formation and evolution of galaxies. Despite the utility of simulations, there often exists a discrepancy between simulated results and observational data that remain not fully understood. In this work we use EAGLE simulations to study fundamental relations, such as the galaxy stellar mass function, the mass-size relation and the mass-metallicity relation. In the simulations the GSMF shows a critical role of aperture size for stellar masses below $10^{11} M_{\odot}$, the consistency of the characteristic mass across different redshifts, and the observed stability of GSMF for masses below $10^{10.5} M_{\odot}$ between redshifts 0 and 1, followed by a decline at higher redshifts. Star-forming galaxies generally exhibit larger sizes, with a slight dependence on the chosen apertures. Star formation and feedback from SN and AGN impact the chemical evolution of galaxies. The star formation also shows an evolution with time, with galaxies at younger stages of the universe exhibiting more significant formation processes. The MZR shows that in EAGLE galaxies, there's an increase in metallicity with stellar mass, except at the high-mass end. Metallicity decreases with higher redshifts, indicating ongoing galaxy evolution. AGN feedback in the simulations have a critical role at the high-mass end of both, the star forming sequence and the mass-metallicity relation.

1. Introduction.

In order to understand how galaxies evolve, cosmological simulations have played a significant role on our understanding of galaxy formation, aiding in the interpretation of observations and the design of new observational tools. Under the current standard cosmological model, galaxies form inside dark-matter halos due to gravitational instability, and grow in mass and size in a bottom-up fashion through mergers (Filipp et al. 2023).

The main physical processes in galaxy formation models include gravity, hydrodynamics and thermal evolution, star formation, black hole formation and growth, star formation feedback, AGN feedback, chemical evolution of stellar populations, and radiative transfer. Gravity shapes the initial structure of galaxies by influencing dark matter halos, which merge and grow over time. Hydrodynamics and thermal evolution determine how gas cools and collapses, forming hot or cold accretion modes. Star formation occurs in dense gas regions, leading to the creation of stars within giant molecular clouds, while black holes grow by accretion of gas (Netzer 2013). Feedback from star formation and AGN activity regulates star formation efficiency by heating and expelling gas (Dekel & Silk 1986; Silk & Rees 1998). Stellar evolution enriches the interstellar medium with heavy elements, influencing cooling rates and star properties. Radiative transfer affects gas heating and the observed properties of galaxies. These processes are often modeled using sub-grid recipes in simulations due to their complexity and varying scales (Somerville & Dave 2014).

Galaxy evolution models in simulations can be broadly categorized into three main types: semi-analytic models, hydrodynamical simulations, and hybrid models (Neistein et al. 2012). Hydrodynamical simulations solve the equations of hydrodynamics alongside gravity, providing detailed, high-resolution models that capture interactions between dark matter, gas, stars, and feedback processes. These simulations produce detailed spatial and temporal information but rely on subgrid models for small-scale processes due to resolution limits, offering detailed insights into physical processes. These simulations collectively

help researchers test theories, interpret observations, and predict the universe's structure and behavior. It is important to note that hydrodynamic models exhibit strengths as well as limitations. Historically, hydrodynamical simulations struggled to match observed galaxy properties, often producing galaxies that were too massive and compact, with early star formation and unquenched star formation in high-mass galaxies. On the other hand hydrodynamical simulations offer advantages, including fewer simplifying assumptions and the ability to evolve dark matter and baryonic components self-consistently. They allow detailed questions and comparisons with more observables, modelling galaxies and the intergalactic medium (IGM) together, crucial for understanding galaxy fueling and feedback cycles. Key to these improvements are subgrid models for supernovae (SN) feedback and AGN feedback, along with advances in resolution and numerical techniques.

In the study of galaxies, several fundamental relationships have been identified that link key galactic properties such as stellar mass, metallicity, star formation rate (SFR), among important others. The galactic stellar mass function (GSMF) serves as a fundamental tool in understanding the distribution of galaxy masses across cosmic time. It reveals insights into the evolution of galaxies and the processes shaping their star formation activity (Weaver et al. 2023). SN feedback impacts star formation rates in low-mass galaxies, particularly those with stellar masses below $5.2 \times 10^9 M_{\odot}$, by reducing the availability of cold gas, while AGN winds become significant in intermediate-mass systems, typically exceeding $10^{9.5} M_{\odot}$, quenching star formation and leading to a minor peak in the GSMF (Scharré et al. 2024). Complementing this, the galaxy mass-size relation (MSR) describes the correlation between a galaxy stellar mass and its physical size and is intimately linked to their morphology, which broadly falls into two categories: spheroidal and disk galaxies. In conjunction with the insights provided by the MSR and the impacts of SN and AGN feedback as described earlier, studying the specific star formation rate (sSFR) serves as a fundamental tool to understand the star formation efficiency, evolutionary paths of galax-

Table 1. Summary of the EAGLE simulations used in this work, adapted from [Crain et al. \(2015\)](#). Columns list: 1) Simulation names, the suffix Ref, Recal and NoAGN correspond to the original subgrid model, a re-calibration and a model without AGN feedback, respectively. 2) The side length of the cubic box, 3) the number of dark matter particles, 4) the baryonic mass resolution 5) the dark matter mass resolution.

Simulation	L [cMpc]	N_{part}	m_{gas} [$M_{\odot}$]	m_{dm} [$M_{\odot}$]
Ref-L100N1504	100	1504^3	0.92	0.13
Ref-L50N0752	50	752^3	0.92	0.13
NoAGN-L50N0752	50	752^3	0.92	0.13
Ref-L25N0752	25	752^3	2.26	0.10
Recal-L25N0752	25	752^3	2.26	0.24

ies, dominant quenching mechanisms, the mass dependence of star formation, and connections to galaxy morphology. By revealing how efficiently galaxies of different masses are forming stars and the underlying processes driving their evolution, such properties provide a comprehensive view of galaxy dynamics and evolution.

The local universe exhibits a strong correlation between stellar mass and elemental abundance (MZR). Comparing the MZR across redshifts is challenging due to selection biases and varying metallicity indicators ([Steidel et al. 2014](#)). The underlying dependencies and evolution of the MZR remain debated. Recent studies suggest the MZR might be a projection of a higher-dimensional relationship involving mass, metallicity, and SFR (e.g. [Ellison et al. 2008](#); [De Rossi et al. 2017](#)). Later on, the fundamental mass-metallicity relation (FMZR) was initially reported by [Mannucci et al. \(2010\)](#) and describes the relation between M^* , metallicity, and SFR.

In the following report, we aim to test and understand this fundamental relationships using EAGLE cosmological simulations ([Crain et al. 2015](#); [Schaye et al. 2015a](#)). Exploring these aspects across different redshifts and simulations, we expected to gain insights into the complex processes shaping the evolution of galaxies.

2. EAGLE simulations.

We are using different cosmological simulations from the EAGLE project. They consist in smoothed particle hydrodynamic (SPH) simulations with different volumes, mass-resolution and subgrid models that reproduces different scenarios for star formation. The simulations are implemented in a Λ CDM cosmology, assuming the parameters from the [Planck Collaboration et al. \(2014\)](#): $\Omega_0 = 0.307$, $\Omega_{\Lambda} = 0.693$, $\Omega_B = 0.048$ and $h = 0.6777$. A more detailed description of the subgrid physics of EAGLE can be found in [Crain et al. \(2015\)](#) and [Schaye et al. \(2015a\)](#). Our sample correspond to central galaxies with masses higher than $10^8 M_{\odot}$ and different redshifts, as mentioned in the following sections.

In [Table 1](#) the main characteristics for the models is summarized. Apart from this, we are using three different subgrid implementations for star formation and AGN feedback. The primary implementation (reference model), a re-calibration of them to better adjust the observational GSMF at higher and lower masses and the mass-size relation and finally a model that do not include AGN feedback. This is deeply described in [Schaye et al. \(2015a\)](#). With the different models we aim to explore the effects of this subgrid implementations on the formation and evolution of galaxies.

3. Results

3.1. Galaxy Mass Function

In [Figure 1](#) we compare the GSMF at different redshifts for Recal-L025N0752 and Ref-L100N1504. For this purpose, we used 0.3 dex bins for the stellar mass for the subhalo (top panel) and a 30pkpc aperture (bottom panel). We can see that the high resolution model, Recal-L025N0752 shows a noisier behaviour because its box size is too small to provide a representative sample. As in [Schaye et al. \(2015b\)](#), for $M_* < 10^{11} M_{\odot}$ the results are insensitive to the aperture. However, for $M_* > 10^{11} M_{\odot}$ the aperture does become important, with larger apertures giving difference in masses of at least 0.3 dex. We know that the GSMF can generally be described approximately by a Schechter function [Schechter \(1976\)](#), i.e. a power law and an exponential break which starts at a characteristic mass, M_{Break} . For both apertures the characteristic mass seems to be conserved for each redshift. We can see nearly no difference in the GSMF for redshifts between 0 and 1 for masses $M_* < 10^{10.5} M_{\odot}$ and then decreases for redshift 2.01 and 3.02. In order to understand if the normalization values are correct it is necessary to compare with observational data. We can see that the GSMF tends to flatten out and become a linear relationship with respect to the redshift as seen in [Furlong et al. \(2015\)](#).

3.2. Mass-Size relation

To study the relation between the stellar mass and the size of the galaxy, in [Figure 2](#) we are comparing the median half-mass radius as a function of the stellar mass. For this purpose we are using bins of 0.5 dex in stellar mass. Both quantities estimated for the reference and the re-calibrated model, with two apertures of 30 and 100 kpc. To separate between the star forming (blue-galaxies) and the quiescent galaxies (red-galaxies), we are considering the color $(g - r) > 0.4$ and $(g - r) < 0.4$, respectively. Other studies use different estimators to separate this two type of galaxies, but the different trend for both types should remain independent of this classification, although with slightly different shapes (e.g. [Shen et al. 2003](#); [de Graaff et al. 2022](#)).

For the reference model at both apertures we can see an increment in size as a function of the stellar mass, as expected. The star-forming galaxies present higher sizes than the quiescent galaxies, with the exception of the high-mass end. This could be explained due to the mass-ranges of the red galaxies, expected to reach higher masses than its counterpart. The difference between the apertures is not relevant for masses lower than $\sim 10^{9.7} M_{\odot}$. At higher masses, smaller apertures result in lower half-mass radius. This can be explained considering that at these high-mass ranges, smaller apertures are not considering the total mass of the galaxy, underestimating its size.

The re-calibrated model shows similar results in terms of the comparison between the apertures, but more pronounced for the red galaxies. The size increment as a function of mass is flatter at lower masses compare to the reference model, until $\sim 10^{9.7}$. After this mass, the increase in size is steeper for the blue galaxies, leading to slightly bigger sizes in these galaxies. The red galaxies instead show smaller sizes. This could be related to the increment in the feedback heating of AGN at higher masses, leading to higher sizes for the blue galaxies due to mechanical AGN winds ([Cochrane et al. 2023](#)).

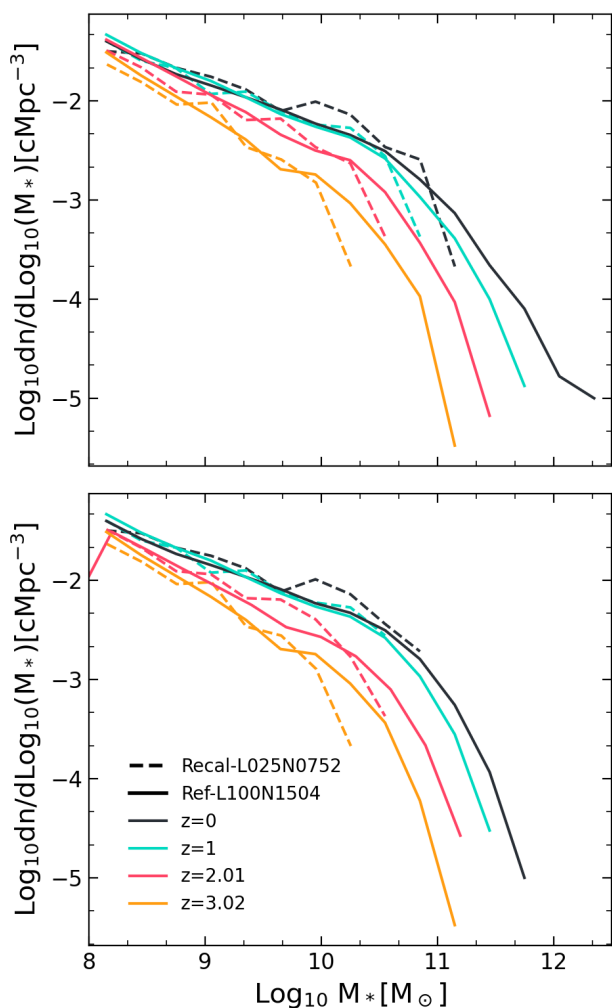


Fig. 1. The GSMF at the redshifts shown in the lower left section of the first panel with solid and dotted lines respectively for simulation Recal-L025N0752 and Ref-L100N1504 and colors differentiate redshifts. Top panel shows the GSMF for the subhalo. Bottom panel shows the GSMF for a 30 pkpc aperture.

3.3. Star Formation and chemical evolution

Implementing different models of star formation and feedback will change the evolution of the galaxy. In principle, the formation rates and the feedback produced by SN and AGN affects also the chemical evolution of galaxies. In this section we aim to study the evolution of the star formation and the gas-phase metallicity of the star forming regions as a function of the stellar mass with a reference model and a No-AGN model.

In Figure 3 we are estimating the median sSFR as a function of different stellar mass bins of size 0.5 dex. This estimation is done for redshift $z = 0, 1, 2.01$ and 3.02 and for two simulations, one with the reference model (dashed line) and another one that is not considering AGN feedback (solid line).

The specific star formation rate is useful to compare the star formation of the galaxies, with respect to its mass. For both models we can see an evolution of this magnitude as a function of time. As expected, at younger stages of the universe galaxies are still in more relevant processes of formation compare to $z = 0$. At the high-mass end there is a slight decrease of the sSFR, that could be explained in different ways. For the reference model, that includes AGN feedback, this decrease is more relevant. We

can expect this due to the heating of the gas via AGN feedback, and only occurs at higher masses because of the necessary mass threshold to host an AGN. These galaxies are also expected to suffer from more mergers than the ones with lower masses, this could also lead to quenching in the No-AGN model (O’Leary et al. 2021).

3.3.1. Mass-Metallicity Relation

In this section we want to understand the dependency between the chemical enrichment of the SF gas regions in galaxies and the stellar mass of them. For this purpose in Figure 4 we are estimating the median metallicity of the star forming gas in bins of 0.5 dex for stellar mass, at the same redshifts and models of Figure 3.

In general, except for the high-mass end, we can see that EAGLE galaxies follow an increasing trend in the metallicity of the SF gas as a function of the stellar mass. Another aspect that holds true for all curves is that for higher redshift the metallicity is lower and the high-mass end moves to the left, suggesting that these galaxies are still evolving and are not fully formed at this times. In particular, for the No-AGN model the increasing trends remains at all redshift, but with different shapes. The increment of the metallicity at the high-mass end is more steeper when we go to higher redshifts. We can compare this with the reference model that includes AGN feedback, this model have a only a steeper increase at $z = 3.02$, compare to lower redshifts of the same model.

As can be seen from the comparison of the model that includes AGN (dashed) and the No-AGN (solid), AGN plays an important role regulating star formation and, as a consequence, the gas enrichment of the star forming regions. We can notice, specially for lower redshifts, that the metallicity is lower for the model that includes AGN. For $z = 3.02$ the difference is still present, but less pronounced. This also suggest that AGN feedback role is more relevant closer to present times.

3.3.2. Relation between AGN feedback and SFR on the MZR

As can be seen from Figure 3 and Figure 4 at higher masses there is a decrease in the star formation and on the slope of the mass-metallicity relation, due to AGN feedback. This suggest a connection between the stellar mass, the star formation and the gas-phase metallicity. There have been previous studies that named this correlation the Fundamental Mass-Metallicity relation (De Rossi et al. 2017, e.g.), but going into much more detail in this is out of the scope of this work. In this section we aim to study only the dispersion in the MZR as a function of the SFR and AGN feedback.

Figure 5 shows the MZR with each galaxy colored by their sSFR. First we can notice that between 10^8 and $10^9 M_\odot$ the No-AGN model have some metal rich galaxies (values higher than 9.4 dex), with high sSFR that does not appear in the reference model. This suggest that the presences of AGN could also play a role at lower masses, but since there are no many galaxies the general trends of the MZR are not significantly affected in this mass range.

The more relevant result from this plot is that the scatter in the MZR is affected by the star formation of the galaxy. Between 10^8 and $10^{10} M_\odot$ the galaxies that have more star formation, tend to fall in the lower regions of the mass-metallicity relation. In contrast with this, the ones with lower star formation tend to have higher metallicities. The meaning of this might be related

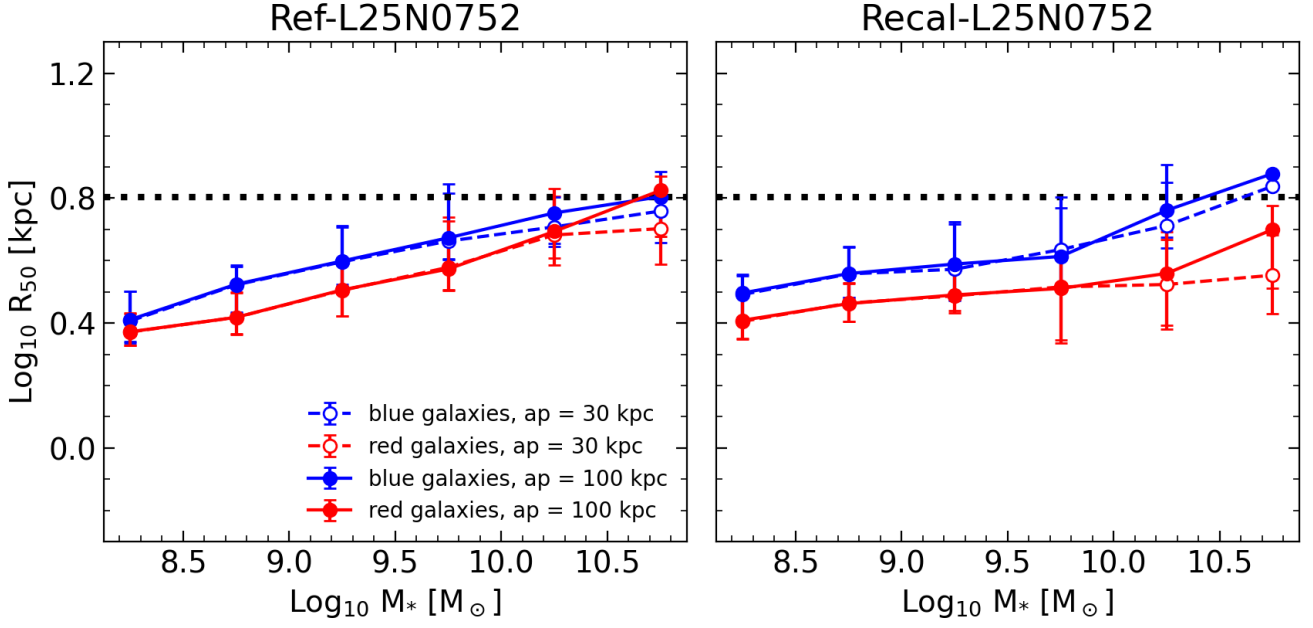


Fig. 2. Stellar Mass-Size relation for the blue and red galaxies of Ref-L25N0752 in the left and Recal-L25N0752 in the right. To estimate the half-mass radius and the stellar masses two apertures were used, 100 kpc (filled dots) and 30 kpc (un-filled). Curves correspond to the median relation and the error bars the 25-th and 75-th percentile in each bin.

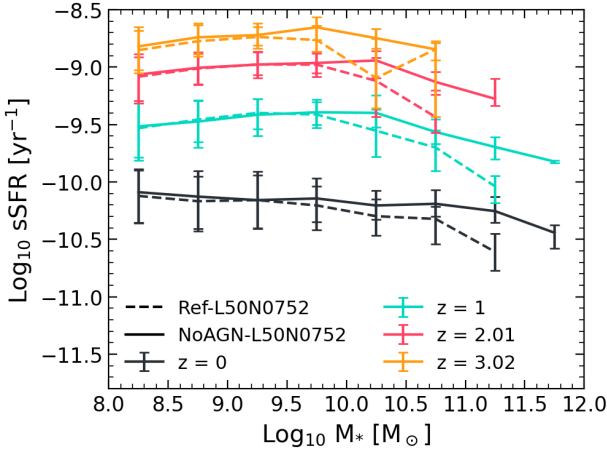


Fig. 3. Relation between the stellar mass and the specific star formation rate for two EAGLE simulations at different redshifts. A reference model that includes AGN (dashed) feedback, and a No AGN simulation (solid). Error bars correspond to 25-th and 75-th percentiles in each bin.

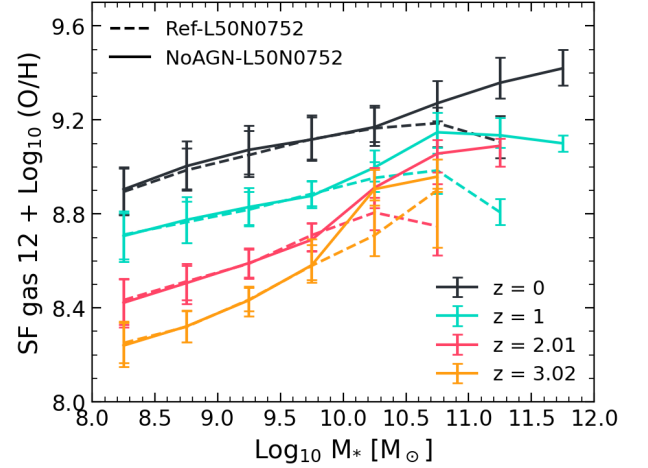


Fig. 4. Relation between the stellar mass and the median gas-phase metallicity of the star forming gas for two EAGLE simulations at different redshifts. A reference model that includes AGN (dashed) feedback, and a No AGN simulation (solid). Error bars correspond to 25-th and 75-th percentiles in each bin.

to the past formation of this galaxies, because those galaxies with lower star formations probably used all the available gas to form stars, getting enriched by this past star formation. In contrast, the gas in higher star formation galaxies is still being enriched by SN. At higher masses we do not see this trend, probably because they already went through the star formation processes that enriched the gas. Instead, for higher masses the more relevant process that shapes the MZR is the AGN feedback.

4. Summary

Using the different EAGLE sub-grid models we study fundamental relations of galaxies. These relations showed, qualitatively, a good agreement with observations and with expected trends. Some of them are summarized in the following:

- i) The analysis of the Galaxy Stellar Mass Function (GSMF) for the models highlights several key points. The Recal-L025N0752 model shows more noise due to its smaller box size. Aperture size is insignificant for stellar masses below $10^{11} M_{\odot}$ but crucial for larger masses. The characteristic mass M_{Break} is consistent across redshifts. The GSMF remains stable for masses below $10^{10.5} M_{\odot}$ between redshifts 0 and 1, but declines at higher redshifts. The GSMF shows a flattening trend with increasing redshift, consistent with previous studies.
- ii) The analysis of the mass-size relation for the models reveals that galaxy size generally increases with stellar mass, consistent across different apertures. Star-forming galaxies are

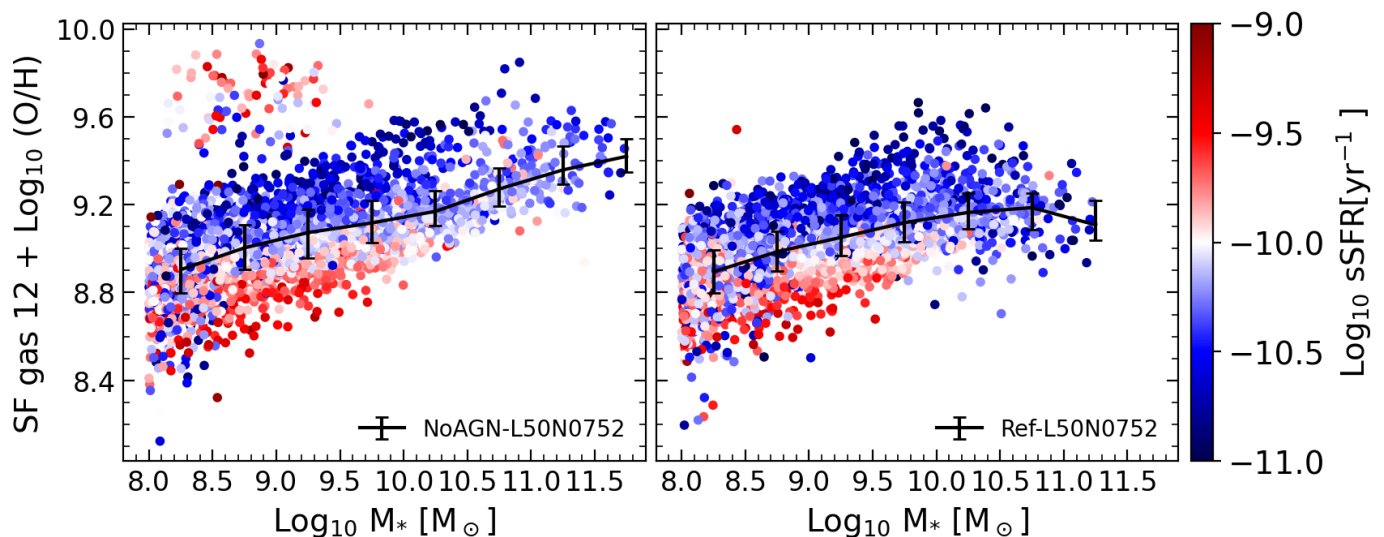


Fig. 5. Relation between the stellar mass and the median gas-phase metallicity of the star forming gas for two EAGLE simulations at $z = 0$. A simulation that do not include AGN feedback processes and a reference model that includes AGN. Error bars correspond to 25-th and 75-th percentiles in each bin. The plot is color-coded by the sSFR of each galaxy, to compare the effects of SFR in the MZR and the dependence with the AGN feedback.

typically larger than quiescent galaxies. The impact of aperture size is minimal for stellar masses below $\sim 10^{9.7} M_{\odot}$, but at higher masses, smaller apertures underestimate the galaxy size. The re-calibrated model shows similar results, but for higher masses result in slightly larger sizes for blue galaxies. This could be attributed to the increase in AGN feedback at this mass range. Leading to larger sizes due to mechanical AGN winds.

- iii) The dependence of the Mass-Metallicity Relation (MZR) and the AGN feedback yields several key findings. Across EAGLE galaxies, there's a general trend of increasing metallicity in star-forming gas as stellar mass increases, except at the high-mass end, indicating ongoing galaxy evolution, particularly at higher redshifts. AGN play a significant role in regulating star formation and gas enrichment, with galaxies featuring AGN feedback exhibiting lower metallicities, especially notable at lower redshifts. This suggests the importance of AGN feedback in shaping the MZR. At higher masses, a decrease in star formation and a flattening of the MZR slope due to AGN feedback suggests a connection between stellar mass, star formation, and gas-phase metallicity. Additionally, the scatter in the MZR is influenced by the specific star formation rate (sSFR) of galaxies, with higher sSFR galaxies tending to have lower metallicities, possibly due to ongoing gas enrichment by supernovae, while lower SFR galaxies tend to have higher metallicities, probably an indicative of past star formation events. In lower mass ranges (10^8 to $10^9 M_{\odot}$), the presence of AGN may contribute to metal-rich galaxies with high sSFR, but this effect is limited due to the small number of galaxies. At higher masses, AGN feedback emerges as the dominant factor shaping the MZR. These conclusions underscore the complex interplay between stellar mass, star formation, AGN feedback, and gas-phase metallicity in shaping the MZR, emphasizing the need for further investigation into these relationships.

- 523, 2409, doi: [10.1093/mnras/stad1528](https://doi.org/10.1093/mnras/stad1528)
 Crain, R. A., Schaye, J., Bower, R. G., et al. 2015, MNRAS, 450, 1937, doi: [10.1093/mnras/stv725](https://doi.org/10.1093/mnras/stv725)
 de Graaff, A., Trayford, J., Franx, M., et al. 2022, MNRAS, 511, 2544, doi: [10.1093/mnras/stab3510](https://doi.org/10.1093/mnras/stab3510)
 De Rossi, M. E., Bower, R. G., Font, A. S., Schaye, J., & Theuns, T. 2017, MNRAS, 472, 3354, doi: [10.1093/mnras/stx2158](https://doi.org/10.1093/mnras/stx2158)
 Dekel, A., & Silk, J. 1986, ApJ, 303, 39, doi: [10.1086/164050](https://doi.org/10.1086/164050)
 Ellison, S. L., Patton, D. R., Simard, L., & McConnell, A. W. 2008, ApJ, 672, L107, doi: [10.1086/527296](https://doi.org/10.1086/527296)
 Filipp, A., Shu, Y., Pakmor, R., Suyu, S. H., & Huang, X. 2023, Astronomy and Astrophysics, 677, A113, doi: [10.1051/0004-6361/202346594](https://doi.org/10.1051/0004-6361/202346594)
 Furlong, M., Bower, R. G., Theuns, T., et al. 2015, Monthly Notices of the Royal Astronomical Society, 450, 4486–4504, doi: [10.1093/mnras/stv852](https://doi.org/10.1093/mnras/stv852)
 Mannucci, F., Cresci, G., Maiolino, R., Marconi, A., & Gnerucci, A. 2010, MNRAS, 408, 2115, doi: [10.1111/j.1365-2966.2010.17291.x](https://doi.org/10.1111/j.1365-2966.2010.17291.x)
 Neistein, E., Khochfar, S., Dalla Vecchia, C., & Schaye, J. 2012, Monthly Notices of the Royal Astronomical Society, 421, 3579–3593, doi: [10.1111/j.1365-2966.2012.20584.x](https://doi.org/10.1111/j.1365-2966.2012.20584.x)
 Netzer, H. 2013, The Physics and Evolution of Active Galactic Nuclei
 O'Leary, J. A., Moster, B. P., Naab, T., & Somerville, R. S. 2021, MNRAS, 501, 3215, doi: [10.1093/mnras/staa3746](https://doi.org/10.1093/mnras/staa3746)
 Planck Collaboration, Ade, P. A. R., Aghanim, N., et al. 2014, A&A, 571, A1, doi: [10.1051/0004-6361/201321529](https://doi.org/10.1051/0004-6361/201321529)
 Scharre, L., Sorini, D., & Davé, R. 2024, arXiv e-prints, arXiv:2404.07252, doi: [10.48550/arXiv.2404.07252](https://doi.org/10.48550/arXiv.2404.07252)
 Schaye, J., Crain, R. A., Bower, R. G., et al. 2015a, MNRAS, 446, 521, doi: [10.1093/mnras/stu2058](https://doi.org/10.1093/mnras/stu2058)
 —. 2015b, MNRAS, 446, 521, doi: [10.1093/mnras/stu2058](https://doi.org/10.1093/mnras/stu2058)
 Schechter, P. 1976, ApJ, 203, 297, doi: [10.1086/154079](https://doi.org/10.1086/154079)
 Shen, S., Mo, H. J., White, S. D. M., et al. 2003, MNRAS, 343, 978, doi: [10.1046/j.1365-8711.2003.06740.x](https://doi.org/10.1046/j.1365-8711.2003.06740.x)
 Silk, J., & Rees, M. J. 1998, A&A, 331, L1, doi: [10.48550/arXiv.astro-ph/9801013](https://doi.org/10.48550/arXiv.astro-ph/9801013)
 Somerville, R., & Dave, R. 2014, Annual Review of Astronomy and Astrophysics, 53, doi: [10.1146/annurev-astro-082812-140951](https://doi.org/10.1146/annurev-astro-082812-140951)
 Steidel, C. C., Rudie, G. C., Strom, A. L., et al. 2014, ApJ, 795, 165, doi: [10.1088/0004-637X/795/2/165](https://doi.org/10.1088/0004-637X/795/2/165)
 Weaver, J. R., Davidzon, I., Toft, S., et al. 2023, Astronomy and Astrophysics, 677, A184, doi: [10.1051/0004-6361/202245581](https://doi.org/10.1051/0004-6361/202245581)

References

Cochrane, R. K., Anglés-Alcázar, D., Mercedes-Feliz, J., et al. 2023, MNRAS,